

24. (a) Obtain an expression for electrostatic potential energy of a system of three charges q , $2q$ and $-3q$ placed at the vertices of an equilateral triangle of side a . 2

OR

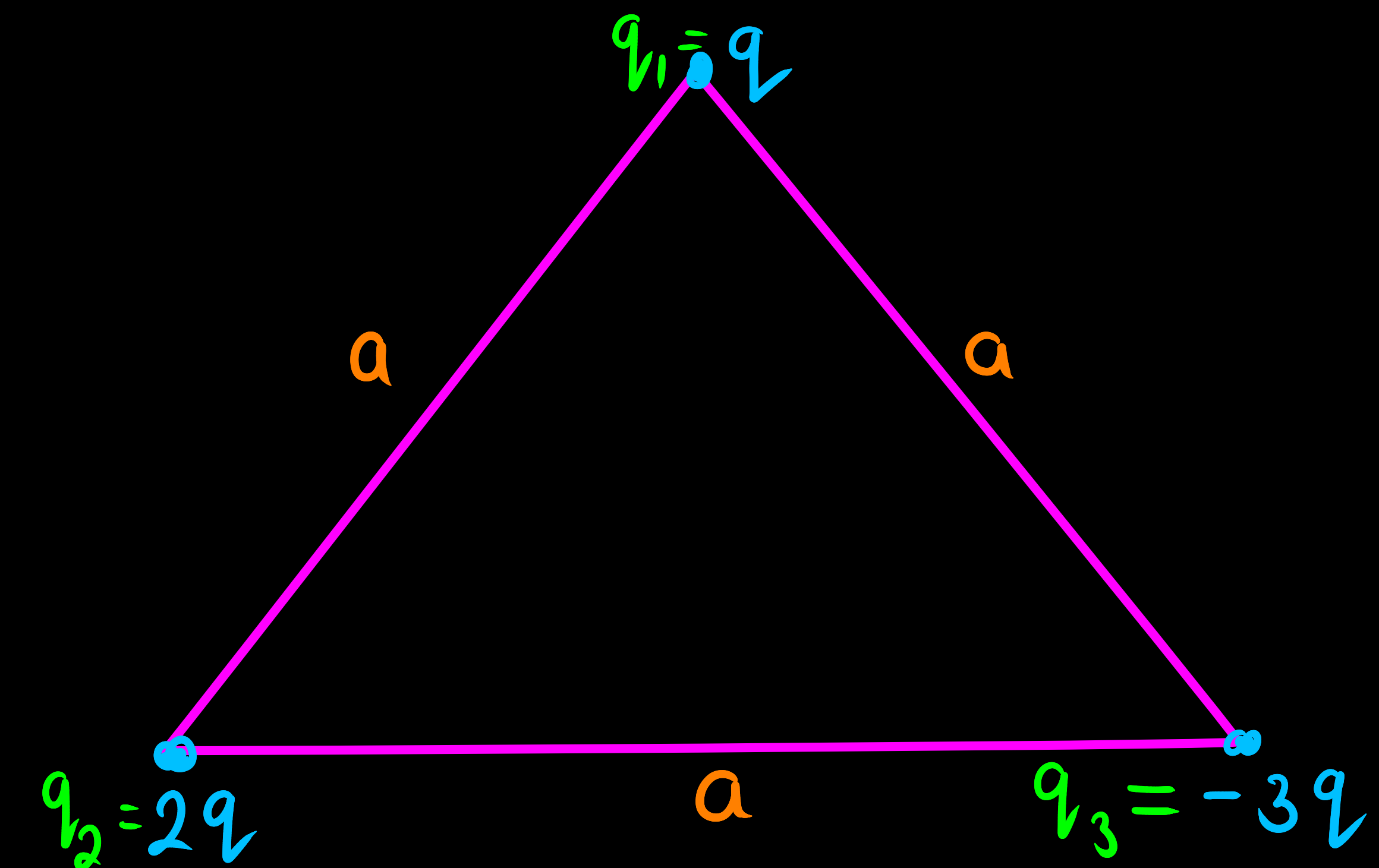
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- (b) Two small conducting balls A and B of radius r_1 and r_2 have charges q_1 and q_2 respectively. They are connected by a wire. Obtain the expression for charges on A and B, in equilibrium. 2

$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right]$$

$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{2q^2}{a} - \frac{3q^2}{a} - \frac{6q^2}{a} \right]$$

$$U = - \frac{1}{4\pi\epsilon_0} \frac{7q^2}{a}$$



24. (a) Obtain an expression for electrostatic potential energy of a system of three charges q , $2q$ and $-3q$ placed at the vertices of an equilateral triangle of side a .

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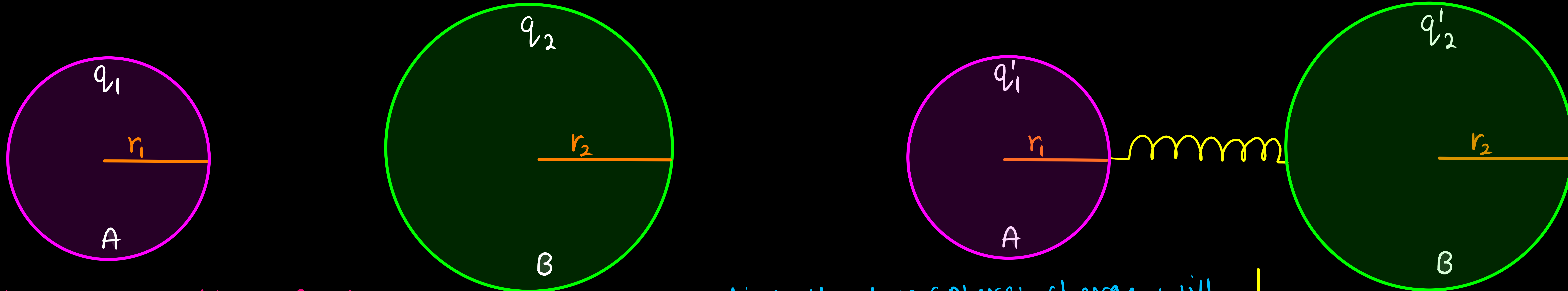
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- (b) Two small conducting balls A and B of radius r_1 and r_2 have charges q_1 and q_2 respectively. They are connected by a wire. Obtain the expression for charges on A and B, in equilibrium.

2

b)



using conservation of charge -

$$\underbrace{q_1' + q_2'}_{\text{final charge}} = \underbrace{q_1 + q_2}_{\text{initial charge}}$$

After connecting the two spheres charge will flow untill they achieve common potential.

$$V_1 = V_2$$

$$K \frac{q_1'}{r_1} = \frac{K q_2'}{r_2}$$

$$r_2 q_1' = r_1 q_2'$$

$$r_2 (q_1 + q_2 - q_2') = r_1 q_2'$$

$$r_2 (q_1 + q_2) - r_2 q_2' = r_1 q_2'$$

$$r_2 (q_1 + q_2) = q_2' (r_1 + r_2)$$

$$q_2' = \frac{(q_1 + q_2) \times r_2}{(r_1 + r_2)}$$

1. Two charges $+q$ each are kept '2a' distance apart. A third charge $-2q$ is placed midway between them. The potential energy of the system is – 1

(A) $\frac{q^2}{8\pi\epsilon_0 a}$

(B) $-\frac{6q^2}{8\pi\epsilon_0 a}$

(C) $\frac{-7q^2}{8\pi\epsilon_0 a}$

(D) $\frac{9q^2}{8\pi\epsilon_0 a}$

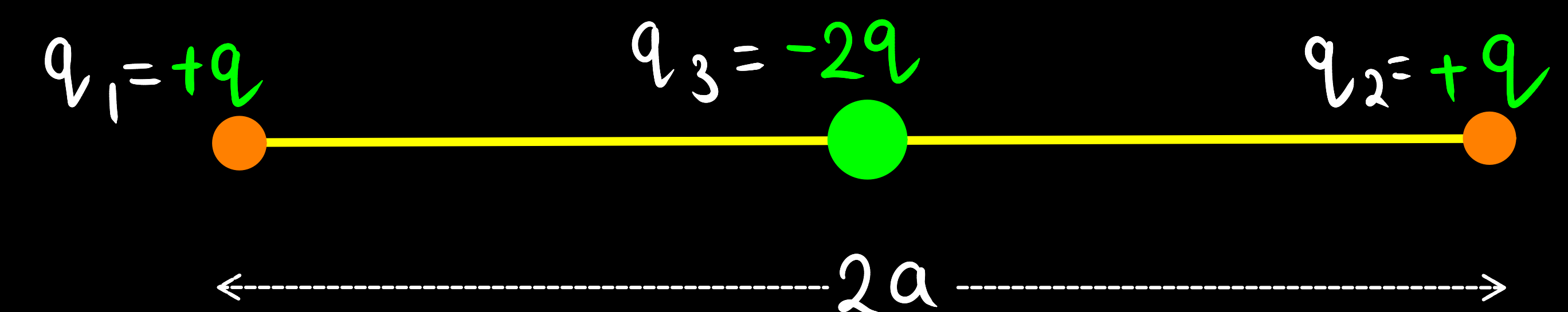
$$U = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right)$$

$$U = \frac{1}{4\pi\epsilon_0} \left(\frac{q^2}{2a} - \frac{2q^2}{a} - \frac{2q^2}{a} \right)$$

$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{q^2}{2a} - \frac{4q^2}{a} \right]$$

$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{q^2}{2a} - \frac{8q^2}{2a} \right]$$

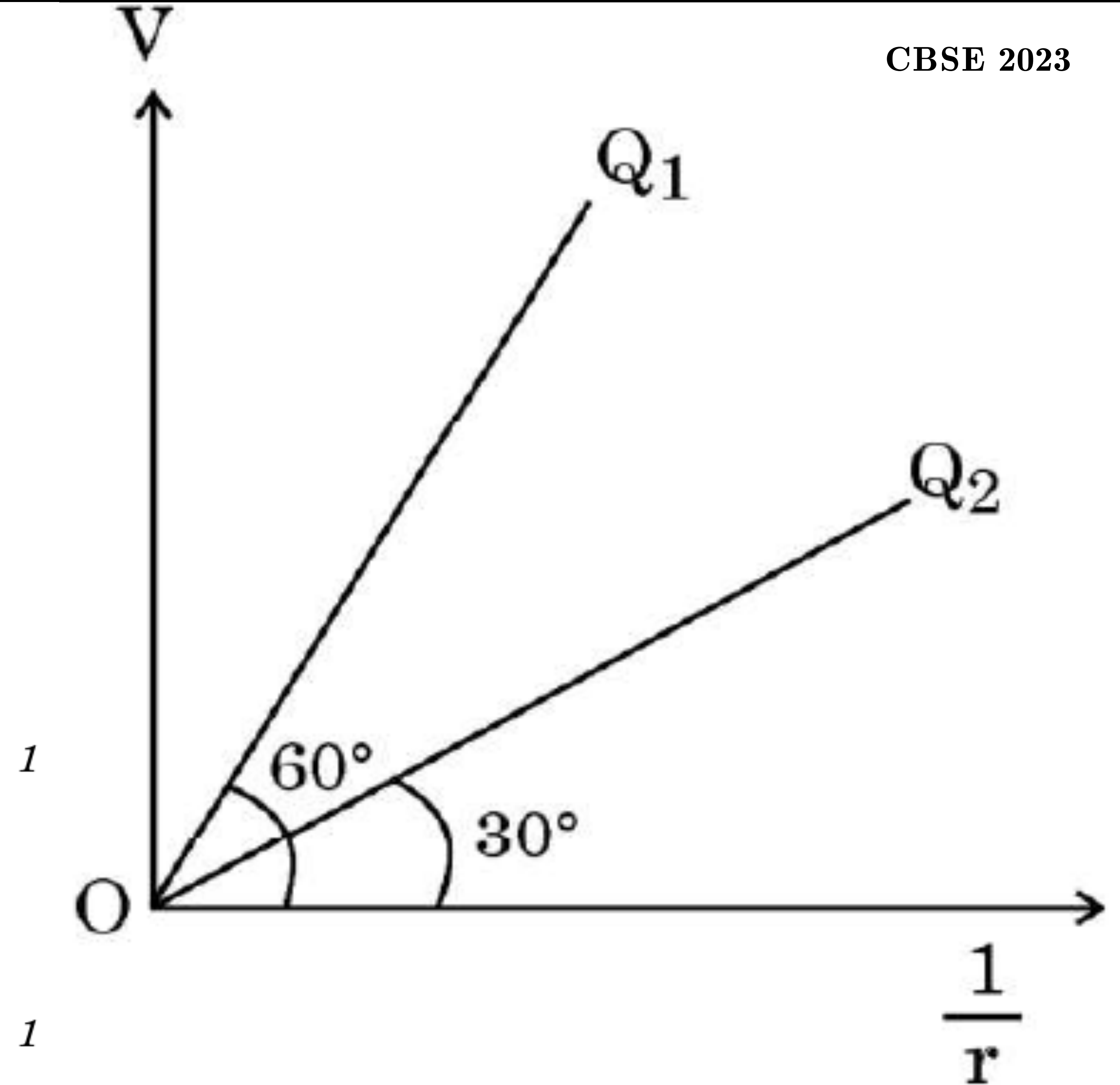
$$U = \frac{1}{4\pi\epsilon_0} \times \frac{(-7q^2)}{2a} = \underline{\underline{\frac{-7q^2}{8\pi\epsilon_0 a}}}$$



34. Electrostatics deals with the study of forces, fields and potentials arising from static charges. Force and electric field, due to a point charge is basically determined by Coulomb's law. For symmetric charge configurations, Gauss's law, which is also based on Coulomb's law, helps us to find the electric field. A charge/a system of charges like a dipole experience a force/torque in an electric field. Work is required to be done to provide a specific orientation to a dipole with respect to an electric field.

Answer the following questions based on the above :

- (a) Consider a uniformly charged thin conducting shell of radius R . Plot a graph showing the variation of $|\vec{E}|$ with distance r from the centre, for points $0 \leq r \leq 3R$. $\rightarrow$ Refer your notebook.
- (b) The figure shows the variation of potential V with $\frac{1}{r}$ for two point charges Q_1 and Q_2 , where V is the potential at a distance r due to a point charge. Find $\frac{Q_1}{Q_2}$.



b) $V = kQ \times \frac{1}{r}$
 $y = \text{Slope}(m) = \tan(\theta)$
 x

$$V_1 = \frac{kQ_1}{r_1}$$

$$\tan \theta_1 = kQ_1 \quad \text{--- (1)}$$

$$V_2 = \frac{kQ_2}{r_2}$$

$$\tan \theta_2 = kQ_2 \quad \text{--- (2)}$$

Divide eqⁿ (1) by (2)

$$\frac{kQ_1}{kQ_2} = \frac{\tan \theta_1}{\tan \theta_2}$$

$$\frac{Q_1}{Q_2} = \frac{\tan 60^\circ}{\tan 30^\circ}$$

$$\frac{Q_1}{Q_2} = \frac{\sqrt{3}}{\frac{1}{\sqrt{3}}}$$

$$\boxed{\frac{Q_1}{Q_2} = 3}$$

c) i) $p = 6 \times 10^{-7} \text{ C-m}$
 $E = 10^4 \text{ N/C}$
 $\theta = 0^\circ$

$$U = -pE \cos \theta$$

$$U = -6 \times 10^{-7} \times 10^4 \times \cos(0)$$

$$U = -6 \times 10^{-3} \text{ J}$$

(c) An electric dipole of dipole moment of $6 \times 10^{-7} \text{ C-m}$ is kept in a uniform electric field of 10^4 N/C such that the dipole moment and the electric field are parallel. Calculate the potential energy of the dipole.

2

OR

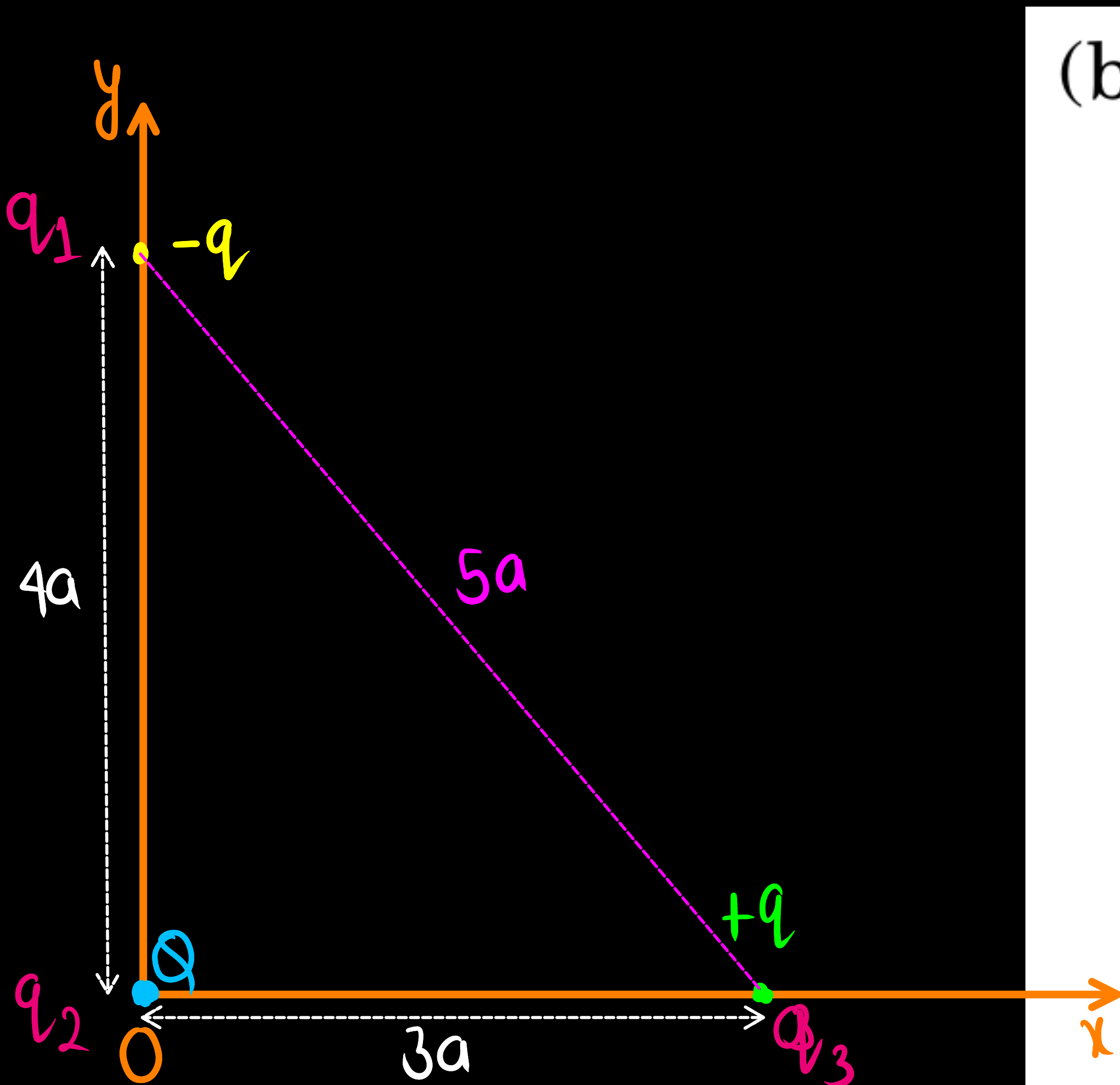
(c) An electric dipole of dipole moment $\vec{p}$ is initially kept in a uniform electric field $\vec{E}$ such that $\vec{p}$ is perpendicular to $\vec{E}$. Find the amount of work done in rotating the dipole to a position at which $\vec{p}$ becomes antiparallel to $\vec{E}$.

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c) ii) $\theta_i = 90^\circ$
 $\theta_f = 180^\circ$
 $\Delta U = -pE (\cos \theta_f - \cos \theta_i)$
 $\Delta U = -pE (\cos 180^\circ - \cos 90^\circ)$
 $\Delta U = -pE (-1 - 0)$

$$\Delta U = pE = W$$



- (b) (i) Define electric potential at a point and write its SI unit.
 $\hookrightarrow \frac{\text{work}}{q}$ $\hookrightarrow \frac{J}{C}$ or volt
- (ii) Two capacitors are connected in series. Derive an expression of the equivalent capacitance of the combination.
 $\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2}$
- (iii) Two point charges $+q$ and $-q$ are located at points $(3a, 0)$ and $(0, 4a)$ respectively in x-y plane. A third charge Q is kept at the origin. Find the value of Q , in terms of q and a , so that the electrostatic potential energy of the system is zero.

Given: $U = 0$

$$\frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right) = 0$$

$$\left(-\frac{qQ}{4a} - \frac{q^2}{5a} + \frac{qQ}{3a} \right) = 0$$

$$\frac{qQ}{3a} - \frac{qQ}{4a} = \frac{q^2}{5a}$$

$$\frac{4qQ - 3qQ}{12a} = \frac{q^2}{5a}$$

$$\frac{qQ}{12a} = \frac{q^2}{5a}$$

$$Q = \frac{12}{5} q$$

a) Motion along x-axis -

$$u_x = 3 \times 10^7 \text{ ms}^{-1}$$

$$x_0 = 0$$

$$a_x = 0 \quad (\because F_x = 0)$$

using $s_x = u_x t + \frac{1}{2} a_x t^2$

$$x - x_0 = u_x t + \frac{1}{2} \times 0 \times t^2$$

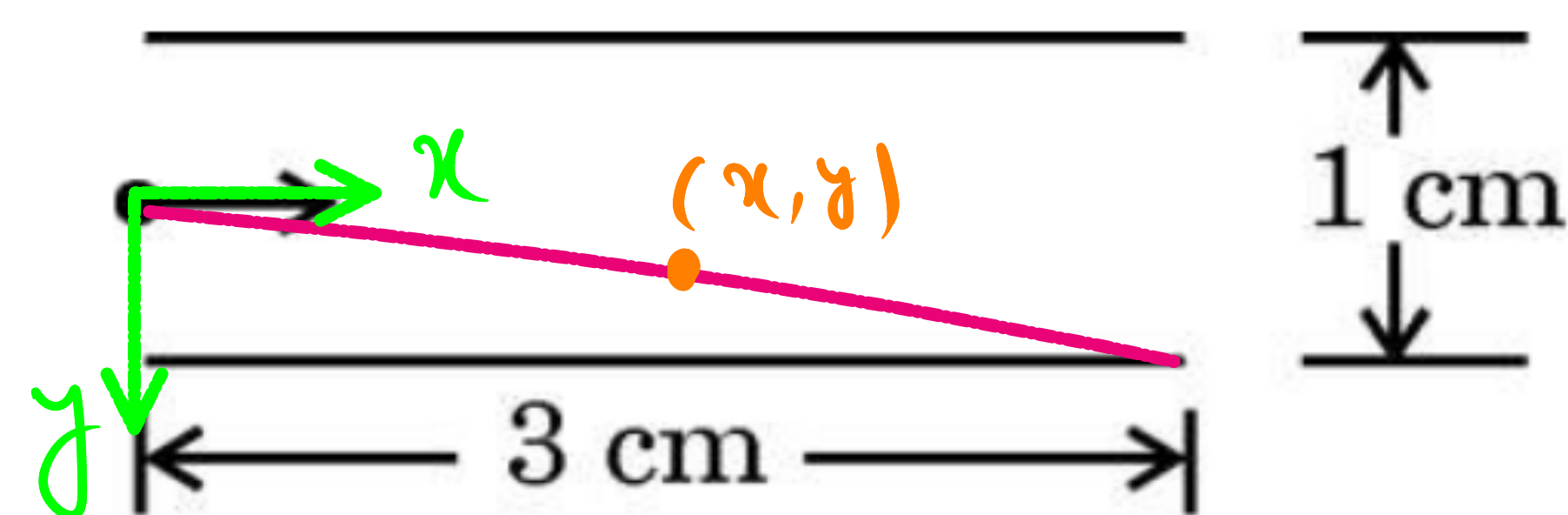
$$x = u_x t$$

$$t = \frac{x}{u_x} \quad \text{--- (i)}$$

$$t = \frac{3 \times 10^{-2} \text{ m}}{3 \times 10^7 \text{ ms}^{-1}}$$

$$t = 10^{-9} \text{ s}$$

34. A beam of electrons moving horizontally with a velocity of $3 \times 10^7 \text{ m/s}$ enters a region between two plates as shown in the figure. A suitable potential difference is applied across the plates such that the electron beam just strikes the edge of the lower plate.



Answer the following questions based on the above :

- (a) How long does an electron take to strike the edge? 1
- (b) What is the shape of the path followed by the electron and why? 1
- (c) Find the potential difference applied. 2

b) Shape of the path - Parabola

$$y = \frac{1}{2} a_y t^2 \quad (\because u_y = 0)$$

$$y = \frac{1}{2} \frac{eE}{m_e} t^2 \quad (\because a_y = \frac{eE}{m_e})$$

$$y = \frac{1}{2} \frac{eE}{m_e} \frac{x^2}{u_x^2} \quad \text{--- (2)}$$

↳ eqⁿ of parabola

When $x = 3 \text{ cm} = 3 \times 10^{-2} \text{ m}$, $y = 0.5 \text{ cm} = 5 \times 10^{-3} \text{ m}$
from eqⁿ (2)

$$5 \times 10^{-3} = \frac{1}{2} \times \frac{1.6 \times 10^{-19} \times E}{9.1 \times 10^{-31}} \times \frac{9 \times 10^{-4}}{9 \times 10^{14}}$$

$$5 \times 10^{-3} = \frac{0.8}{9.1} \times 10^{-6} \times E$$

$$E = \frac{5 \times 9.1}{0.8} \times 10^3$$

$$E = 56.875 \times 10^3 \frac{\text{V}}{\text{m}}$$

$$V = Ed$$

$$V = 56.875 \times 10^3 \frac{\text{V}}{\text{m}} \times 10^{-2} \text{ m}$$

$$V = 568.75 \text{ V}$$

28. Three point charges Q_1 ($-15 \mu\text{C}$), Q_2 ($10 \mu\text{C}$) and Q_3 ($16 \mu\text{C}$) are located at (0 cm, 0 cm), (0 cm, 3 cm) and (4 cm, 3 cm) respectively. Calculate the electrostatic potential energy of this system of charges.

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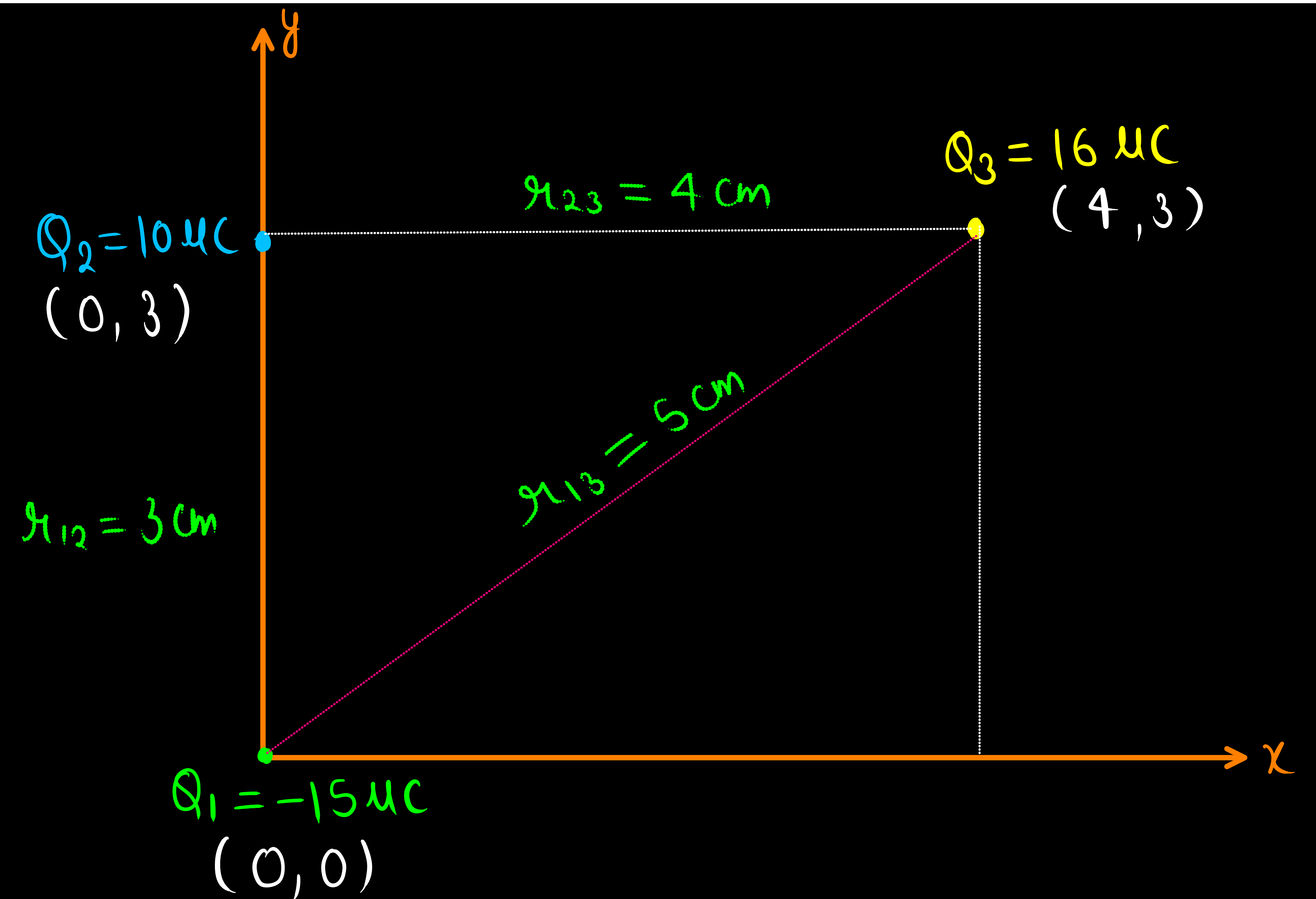
$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{Q_1 Q_2}{r_{12}} + \frac{Q_1 Q_3}{r_{13}} + \frac{Q_2 Q_3}{r_{23}} \right]$$

$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{-15 \times 10^{-6} \times 10 \times 10^{-6}}{3 \times 10^{-2}} - \frac{15 \times 10^{-6} \times 16 \times 10^{-6}}{5 \times 10^{-2}} + \frac{10 \times 10^{-6} \times 16 \times 10^{-6}}{4 \times 10^{-2}} \right]$$

$$U = \frac{9 \times 10^9 \times 10^{-12}}{10^{-2}} [-50 - 48 + 40]$$

$$U = 9 \times 10^{-1} \times (-58)$$

$$U = 52.2 \text{ J}$$



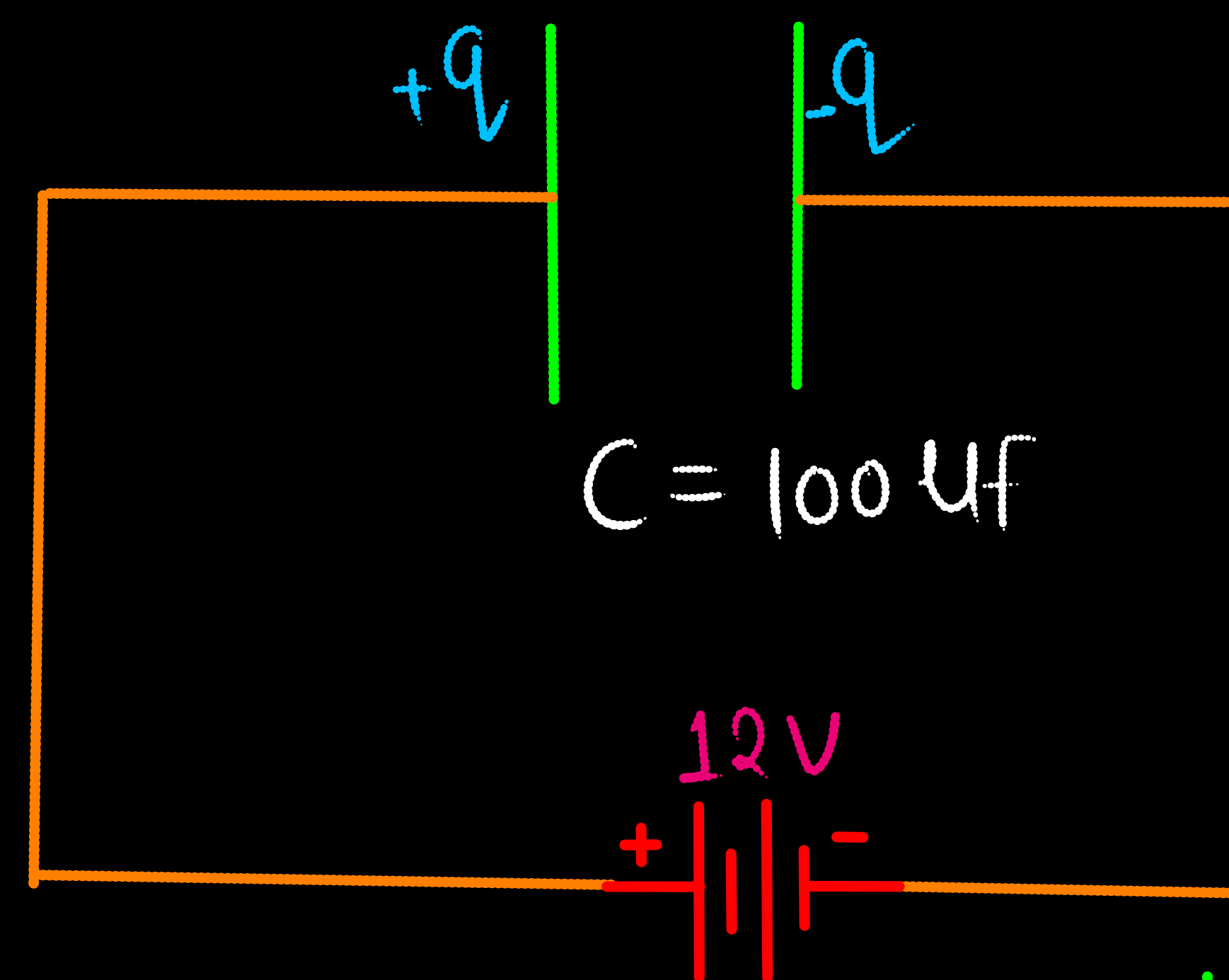
a) $C = 100 \mu\text{F}$
 $V = 12 \text{ V}$
 $U = \frac{1}{2} C V^2$
 $U = \frac{1}{2} \times 100 \times 10^{-6} \times 12^2$
 $U = 72 \times 10^{-4} \text{ J}$

b) $q_1' + q_2' = q$
 Also, $V_1 = V_2$
 $\frac{q_1'}{C} = \frac{q_2'}{C}$
 $q_1' = q_2' = \frac{q}{2}$
 $V' = \frac{q_1'}{C} = \frac{q}{2C} = \frac{V}{2}$
 $C' = C + C = 2C$

30. A $100 \mu\text{F}$ capacitor is charged by a 12 V battery.

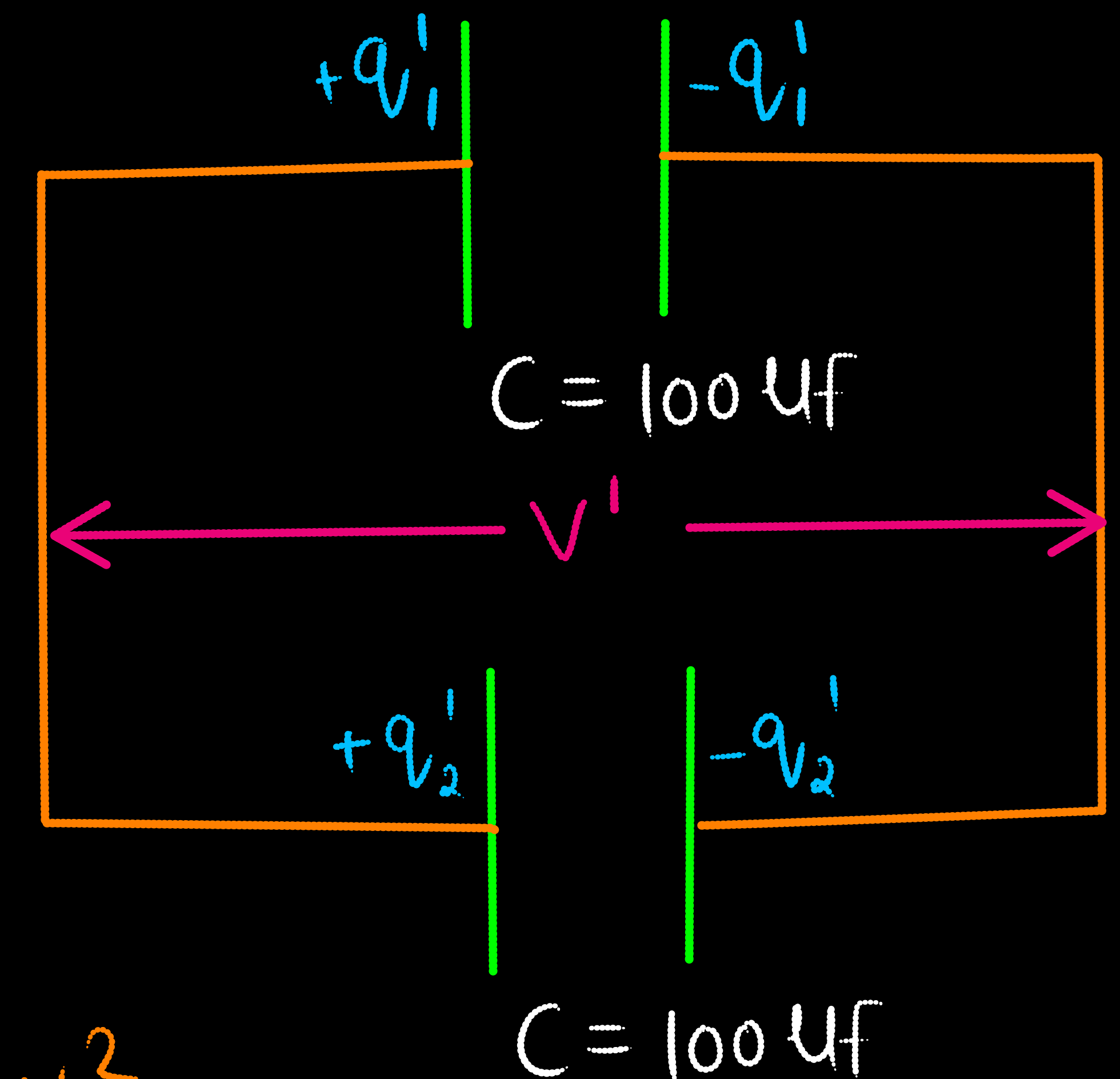
- (a) How much electrostatic energy is stored by the capacitor ?
 (b) The capacitor is disconnected from the battery and connected in parallel to another uncharged $100 \mu\text{F}$ capacitor. What is the electrostatic energy stored by the system ?

3



$$U' = \frac{1}{2} C' V'^2$$

$$U' = \frac{1}{2} \times 2C \times \frac{V^2}{4}$$



$$U' = \frac{1}{2} \times \frac{1}{2} C V^2$$

$$U' = \frac{1}{2} U = 36 \times 10^{-4} \text{ J}$$